

assignment_6_solutions_rubric

November 17, 2023

0.1 Problem 1a: Install COSMIC and run the first example; then print the bpp with the tphys, mass_1, mass_2, kstar_1, kstar_2, porb, ecc, and evol_type columns

6 points for installation (guaranteed if the rest of the assignment is done)

3 points for running the binary and printing

3 points for words description

```
[ ]: from cosmic.sample.initialbinarytable import InitialBinaryTable
from cosmic.evolve import Evolve
import matplotlib.pyplot as plt
import numpy as np

from cosmic.plotting import evolve_and_plot

[ ]: single_binary = InitialBinaryTable.InitialBinaries(m1=85.543645, m2=84.99784,
↳porb=446.795757, ecc=0.448872, tphysf=13700.0, kstar1=1, kstar2=1,
↳metallicity=0.002)

[ ]: BSEDdict = {'xi': 1.0, 'bhflag': 1, 'neta': 0.5, 'windflag': 3, 'wdflag': 1,
↳'alpha1': 1.0, 'pts1': 0.001, 'pts3': 0.02, 'pts2': 0.01, 'epsnov': 0.001,
↳'hewind': 0.5, 'ck': 1000, 'bwind': 0.0, 'lambdaf': 0.0, 'mxns': 3.0, 'beta':
↳-1.0, 'tflag': 1, 'acc2': 1.5, 'grflag': 1, 'remnantflag': 4, 'ceflag': 0,
↳'eddfac': 1.0, 'ifflag': 0, 'bconst': 3000, 'sigma': 265.0, 'gamma': -2.0,
↳'pism': 45.0, 'natal_kick_array' : [[-100.0,-100.0,-100.0,-100.0,0.0], [-100.
↳0,-100.0,-100.0,-100.0,0.0]], 'bhsigmafrac' : 1.0, 'polar_kick_angle' : 90,
↳'qcrit_array' : [0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0,0.0],
↳'cekickflag' : 2, 'cehstarflag' : 0, 'cemerageflag' : 0, 'ecsn' : 2.
↳25, 'ecsn_mlow' : 1.6, 'aic' : 1, 'ussn' : 0, 'sigmadiv' :-20.0, 'qcflag' :
↳1, 'eddlimflag' : 0, 'fprimc_array' : [2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.
↳0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0,2.0/21.0],
↳'bhspinflag' : 0, 'bhspinmag' : 0.0,
↳'rejuv_fac' : 1.0, 'rejuvflag' : 0, 'htpmb' : 1, 'ST_cr' : 1, 'ST_tide' : 1,
↳'bdecayfac' : 1, 'rembar_massloss' : 0.5, 'kickflag' : 0, 'zsun' : 0.014,
↳'bhms_coll_flag' : 0, 'don_lim' : -1, 'acc_lim' : -1, 'rtmsflag' : 0}
```

```
[ ]: bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,
↳ BSEDict=BSEDict)
```

```
[ ]: print(bpp[['tphys', 'mass_1', 'mass_2', 'kstar_1', 'kstar_2', 'porb', 'ecc',
↳ 'evol_type']])
```

	tphys	mass_1	mass_2	kstar_1	kstar_2	porb \
0	0.000000	85.543645	84.997840	1.0	1.0	446.795757
0	3.717075	72.720332	72.376321	2.0	1.0	617.238959
0	3.718373	72.589218	72.377845	2.0	1.0	252.923207
0	3.720012	71.959504	72.806393	4.0	1.0	252.533894
0	3.740038	32.352601	110.573279	4.0	1.0	628.743501
0	3.741215	32.176152	110.618802	7.0	1.0	628.922011
0	4.071374	25.488585	106.891756	7.0	1.0	731.786530
0	4.071374	24.988585	106.891756	14.0	1.0	737.361815
0	4.894369	24.988590	88.885767	14.0	2.0	988.424464
0	4.895887	24.989628	88.681512	14.0	2.0	986.044004
0	4.896841	24.990676	88.469775	14.0	4.0	986.208557
0	4.896841	24.990676	88.469775	14.0	4.0	986.208557
0	4.896841	24.990676	41.981621	14.0	7.0	81.319009
0	4.896841	24.990676	41.981621	14.0	7.0	81.319009
0	5.205274	24.990687	31.554344	14.0	7.0	114.082486
0	5.205274	24.990687	31.054344	14.0	14.0	116.140982
0	13700.000000	24.990687	31.054344	14.0	14.0	116.033619

	ecc	evol_type
0	0.448872	1.0
0	0.448872	2.0
0	0.000000	3.0
0	0.000000	2.0
0	0.000000	4.0
0	0.000000	2.0
0	0.000000	15.0
0	0.003791	2.0
0	0.003791	2.0
0	0.000000	3.0
0	0.000000	2.0
0	0.000000	7.0
0	0.000000	8.0
0	0.000000	4.0
0	0.000000	16.0
0	0.008921	2.0
0	0.008921	10.0

0.2 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 3.7 Myr. The primary fills its Roche lobe while it is on the Hertzsprung Gap and the mass transfer

remains stable transferring over 40 solar masses of material to the companion. This results in the binary having a wider orbit than it started out with. Eventually, the primary envelope is fully stripped, then the primary forms a black hole with 25 solar masses. The formation of the black hole doesn't change the orbit very much, so the natal kick is probably small. The secondary then evolves off the main sequence at 4.9 Myr and fills its Roche lobe on the Hertzsprung Gap. Once the secondary evolves to core helium burning, the mass transfer becomes unstable and leads to a common envelope evolution which produces a binary with an 81 day orbital period. The black hole formed by the primary doesn't accrete much material. The secondary forms a 31 solar mass black hole and the orbital period remains at 116 days even at a Hubble time of evolution.

0.5 pt off for missing Roche lobe overflow statement (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes

0.5 pt off for not describing how the masses change

0.3 Problem 1b: Evolve a ZAMS binary with

0.3.1 $mass_1 = 6.0 M_{\odot}$, $mass_2 = 1.5 M_{\odot}$, $porb = 600$ days, $ecc = 0$, $Z = 0.014$

3 points for running the binary and printing

3 points for orbital period plot

3 points for mass plot

3 points for words description

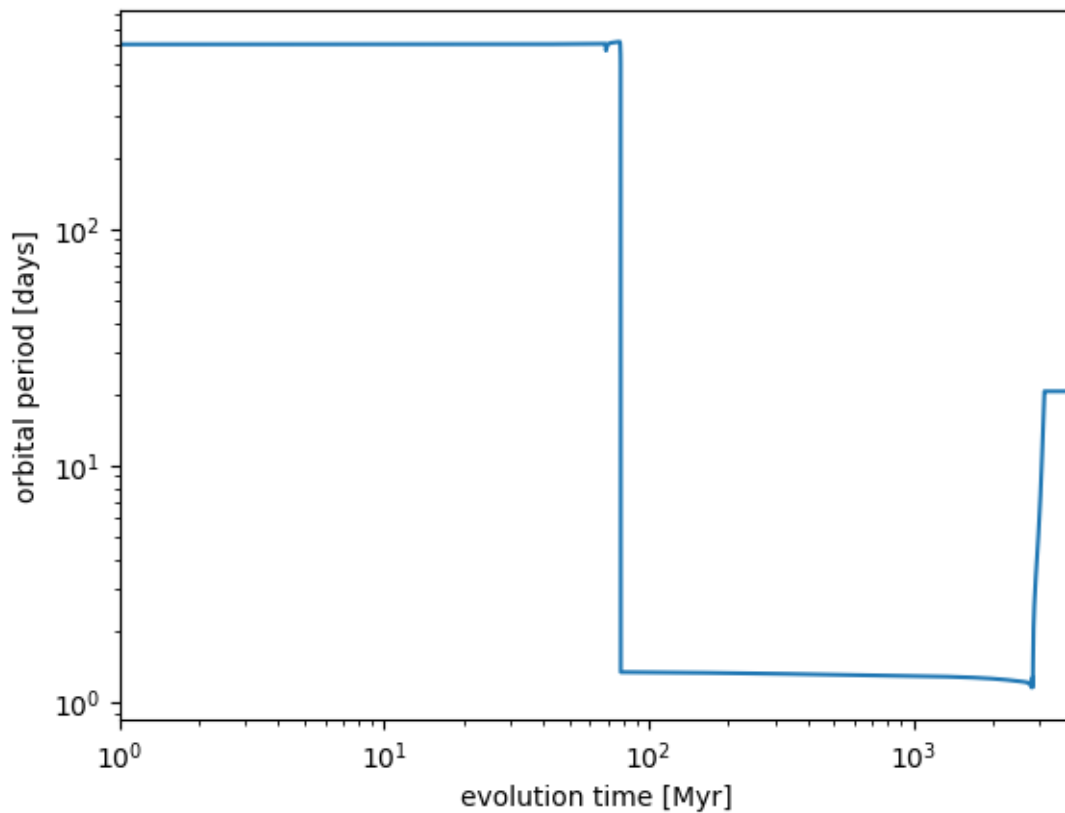
```
[ ]: single_binary = InitialBinaryTable.InitialBinaries(m1=6.0, m2=1.5, porb=600.0,
    ↪ecc=0.0, tphysf=13700.0,
    kstar1=1, kstar2=1,
    ↪metallicity=0.014)
bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,
    ↪BSEDict=BSedict, dtp=0.0)
bpp[['tphys', 'kstar_1', 'kstar_2', 'mass_1', 'mass_2', 'porb', 'evol_type']]
```

```
[ ]:      tphys  kstar_1  kstar_2   mass_1   mass_2      porb  evol_type
0      0.000000      1.0      1.0  6.000000  1.500000  600.000000      1.0
0      68.559582      2.0      1.0  5.979993  1.500000  603.213995      2.0
0      68.809728      3.0      1.0  5.979524  1.500001  603.349846      2.0
0      68.918809      4.0      1.0  5.978294  1.500014  578.837101      2.0
0      78.038656      5.0      1.0  5.890052  1.500761  600.803163      2.0
0      78.347085      5.0      1.0  5.875955  1.501061  489.380684      3.0
0      78.347085      5.0      1.0  5.875955  1.501061  489.380684      7.0
0      78.347085      8.0      1.0  1.493853  1.501061   2.964431      8.0
0      78.347085      8.0      1.0  1.493853  1.501061   2.964431      7.0
0      78.347085     11.0      1.0  0.932782  1.501061   1.339521      8.0
0      78.347085     11.0      1.0  0.932782  1.501061   1.339521      4.0
```

0	2723.537548	11.0	2.0	0.932782	1.501061	1.201748	2.0
0	2725.324049	11.0	2.0	0.932783	1.501055	1.205693	3.0
0	2868.091958	11.0	3.0	0.933814	0.560470	2.729554	2.0
0	3129.365737	11.0	3.0	0.942573	0.227356	20.542647	4.0
0	3139.028907	11.0	10.0	0.942784	0.226925	20.546670	2.0
0	13700.000000	11.0	10.0	0.942784	0.226925	20.546565	10.0

```
[ ]: plt.plot(bcm.tphys, bcm.porb)
plt.xlim(1, 4000)
plt.xscale('log')
plt.yscale('log')
plt.ylabel('orbital period [days]')
plt.xlabel('evolution time [Myr]')
```

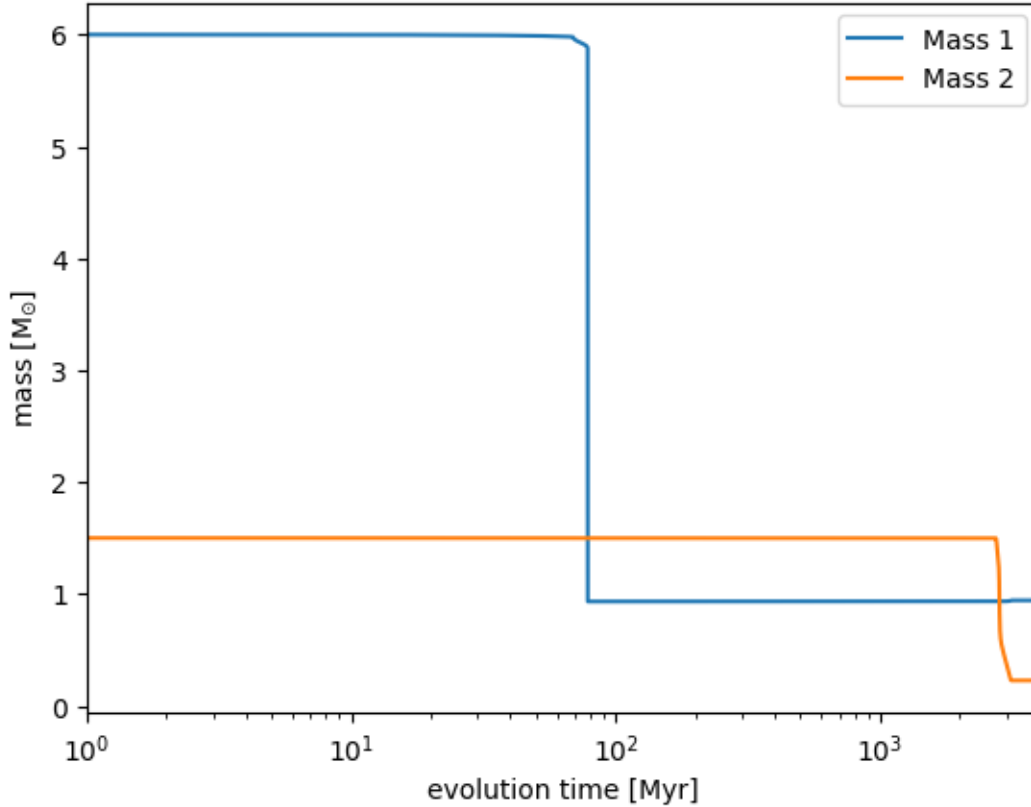
```
[ ]: Text(0.5, 0, 'evolution time [Myr]')
```



```
[ ]: plt.plot(bcm.tphys, bcm.mass_1, label='Mass 1')
plt.plot(bcm.tphys, bcm.mass_2, label='Mass 2')
plt.xlim(1, 4000)
plt.xscale('log')
plt.legend()
```

```
plt.xlabel('evolution time [Myr]')
plt.ylabel('mass [M$_{\odot}$]')
```

```
[ ]: Text(0, 0.5, 'mass [M$_{\odot}$]')
```



```
[ ]:
```

0.4 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 68.6 Myr and subsequently evolves across the Hertsprung Gap, climbs the red giant branch, and begins core helium burning. The primary fills its Roche lobe while it is on the early asymptotic giant branch and the mass transfer becomes unstable. This results in a common envelope evolution that leaves behind a much shorter orbit. Since the Roche overflow resulted in a common envelope, the mass of the secondary didn't increase at all, but the mass of the primary was significantly decreased. At ~2.6 Gyr the secondary evolves off the main sequence and fills it's Roche lobe while it is crossing the Hertzsprung Gap. The mass transfer remains stable, but the WD can't accrete much material, so there is little mass exchange and the secondary just loses mass. There is a kink in the time evolution of the secondary because the secondary starts to ascend the giant branch. The orbital period initially decreases, but then increases because there was a mass ratio inversion and the secondary mass changes from being more massive than the primary to being less massive than

the primary. The end stage is a Carbon-Oxygen and Helium white dwarf binary with an orbital period of 20 days.

0.5 pt off for missing Roche lobe overflow statement going to CE for primary and Stable for secondary (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes

0.5 pt off for not describing how the masses change

0.5 Problem 1c: Evolve a ZAMS binary with a random seed of 5 and

0.5.1 $mass_1 = 50.0 M_{\odot}$, $mass_2 = 35 M_{\odot}$, $porb = 120$ days, $ecc = 0$, $Z = 0.014$

3 points for running the binary and printing

3 points for orbital period plot

3 points for mass plot

3 points for words description

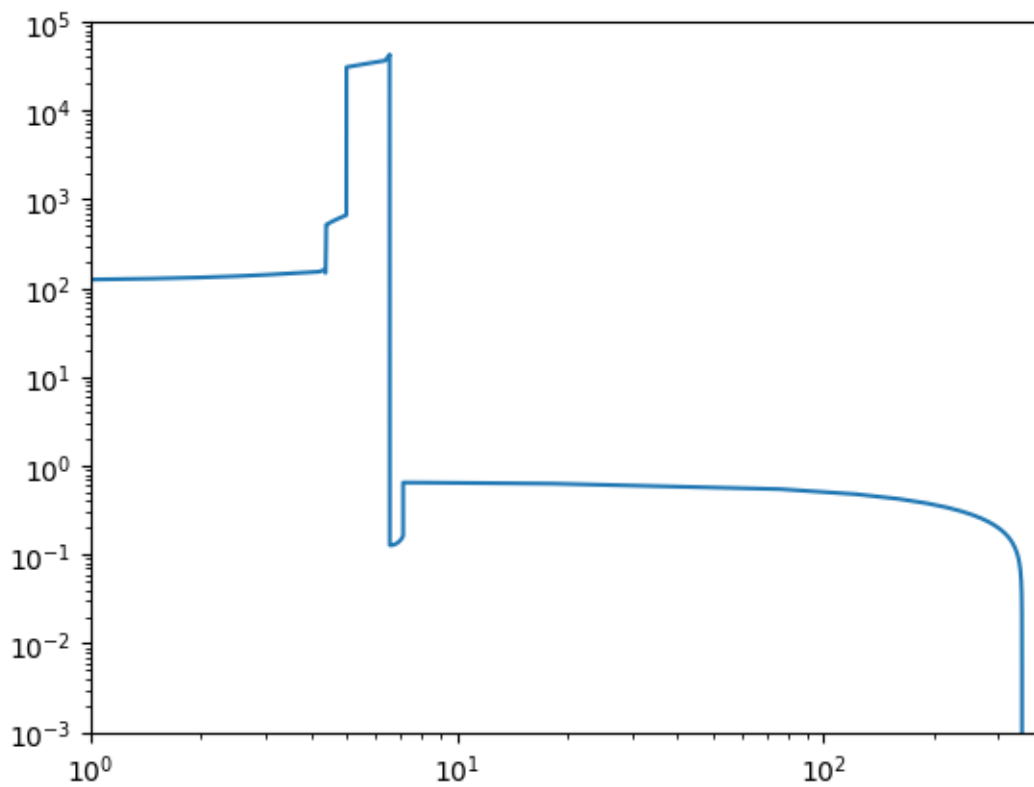
```
[ ]: np.random.seed(5)
single_binary = InitialBinaryTable.InitialBinaries(m1=50.0, m2=35.0, porb=120.
    ↪0, ecc=0.0, tphysf=13700.0,
                                     kstar1=1, kstar2=1,
    ↪metallicity=0.014)
bpp, bcm, initC, kick_info = Evolve.evolve(initialbinarytable=single_binary,
    ↪BSEDict=BSEDict, dtp=0.0)
print(bpp[['tphys', 'mass_1', 'mass_2', 'kstar_1', 'kstar_2', 'porb', 'ecc',
    ↪'evol_type']].round(3))
```

	tphys	mass_1	mass_2	kstar_1	kstar_2	porb	ecc	evol_type
0	0.000	50.000	35.000	1.0	1.0	120.000	0.000	1.0
0	4.358	40.464	32.495	2.0	1.0	162.882	0.000	2.0
0	4.361	40.436	32.495	2.0	1.0	162.923	0.000	3.0
0	4.364	40.032	32.854	4.0	1.0	161.846	0.000	2.0
0	4.392	14.802	57.774	4.0	1.0	521.423	0.000	4.0
0	4.397	14.736	57.803	7.0	1.0	520.661	0.000	2.0
0	4.954	8.846	55.665	8.0	1.0	658.361	0.000	2.0
0	4.984	8.554	55.544	8.0	1.0	666.877	0.000	15.0
0	4.984	4.180	55.544	14.0	1.0	30894.690	0.921	2.0
0	6.538	4.180	46.884	14.0	2.0	42227.385	0.921	2.0
0	6.542	4.180	46.486	14.0	2.0	956.646	0.000	3.0
0	6.542	4.180	46.486	14.0	2.0	956.646	0.000	7.0
0	6.542	4.180	17.623	14.0	7.0	0.158	0.000	8.0
0	6.542	4.180	17.623	14.0	7.0	0.158	0.000	4.0
0	7.095	4.269	9.562	14.0	8.0	0.157	0.000	2.0
0	7.123	4.275	9.248	14.0	8.0	0.165	0.000	16.0

0	7.123	4.275	4.984	14.0	14.0	0.649	0.593	2.0
0	348.667	4.275	4.984	14.0	14.0	0.000	0.000	3.0
0	348.667	9.259	0.000	14.0	15.0	0.000	-1.000	6.0
0	13700.000	9.259	0.000	14.0	15.0	0.000	-1.000	10.0

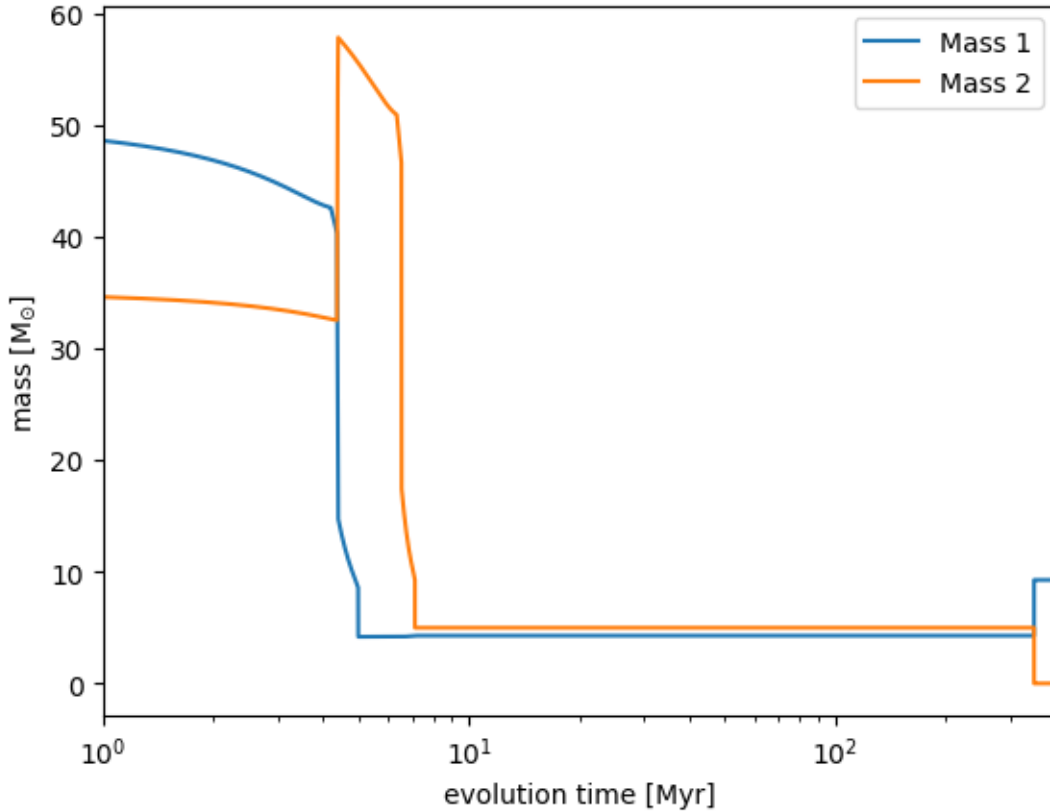
```
[ ]: plt.plot(bcm.tphys, bcm.porb)
plt.yscale('log')
plt.xscale('log')
plt.xlim(1, 400)
plt.ylim(1e-3, 1e5)
```

```
[ ]: (0.001, 100000.0)
```



```
[ ]: plt.plot(bcm.tphys, bcm.mass_1, label='Mass 1')
plt.plot(bcm.tphys, bcm.mass_2, label='Mass 2')
plt.xlim(1, 400)
plt.xscale('log')
plt.legend()
plt.xlabel('evolution time [Myr]')
plt.ylabel('mass [M$_{\odot}$]')
```

```
[ ]: Text(0, 0.5, 'mass [M$_{\odot}$]')
```



[]:

[]:

0.6 Description:

The binary starts out with two main sequence stars, then the primary evolves off the main sequence at 4.358 Myr. The primary fills its Roche lobe while it is on the Hertzsprung Gap and the mass transfer remains stable transferring 35 solar masses of material to the companion. This results in the binary having a wider orbit than it started out with. Eventually, the primary envelope is fully stripped, and the left over helium star loses mass from stellar winds which further widens the orbit. The primary forms a black hole with 4 solar masses at 4.99 Myr. The formation of the black hole changes the orbit significantly by both widening and increasing the eccentricity, so the natal kick is probably strong. The secondary then evolves off the main sequence at 6.5 Myr and fills its Roche lobe on the Hertzsprung Gap and immediately leads to a common envelope evolution which produces a binary with an 0.16 day orbital period. The black hole formed by the primary doesn't accrete during the common envelope. The secondary forms a 4.98 solar mass black hole at 7.1 Myr and the orbital period increases to 0.35 days. The two black holes merge at 349 Myr. The main differences between this evolution and the evolution in 1a are the different black hole masses and the fact that the black holes in 1a don't merge in a Hubble time, but this black hole would merge and produce gravitational waves.

0.5 pt off for missing Roche lobe overflow statement (stable for primary and CE for secondary) (max of 1 pt off for missing both RLOs)

0.5 pt off for not describing how the orbit changes (especially due to kicks)

0.5 pt off for not describing how the masses change

0.5 pt off for not comparing to the evolution of 1a.

0.7 Problem 2, 6 pts total

```
[ ]: import astropy.units as u
import astropy.constants as c
```

The strain is

$$h^2(\mathcal{M}_c, f_{\text{GW}}, D) = \frac{128}{5} \frac{(G\mathcal{M}_c)^{10/3}}{c^8 D^2} (\pi f_{\text{GW}})^{4/3}$$

and the chirp is

$$\dot{f}(\mathcal{M}_c, f_{\text{GW}}) = \frac{96}{5\pi} \frac{(G\mathcal{M}_c)^{5/3}}{c^5} (2\pi f_{\text{orb}})^{11/3}$$

where $f_{\text{GW}} = 2 f_{\text{orb}}$.

We can solve for the chirp mass from $\dot{f}$ as

$$\mathcal{M}_c = \left(\frac{96}{5} \frac{\dot{f}}{(2\pi f_{\text{orb}})^{11/3}} \right)^{3/5} \frac{c^3}{G}$$

and for the distance as

$$D = \sqrt{\frac{128}{5} \frac{(G\mathcal{M}_c)^{10/3}}{c^8 h^2} (\pi f_{\text{GW}})^{4/3}}$$

```
[ ]: def chirp_mass_from_chirp(f_dot, f_orb):
    return ((96/5) * f_dot/(2 * np.pi * f_orb)**(11/3))**(3/5) * c.c**3 / c.G

def distance_from_strain_chirp_mass(strain, chirp_mass, f_gw):
    dist = np.sqrt(128/5 / strain**2 * (np.pi * f_gw)**(4/3) * (c.G *
↪chirp_mass)**(10/3)/c.c**8 )
    return dist
```

```
[ ]: fdot = 1.1e-16 * (u.s)**(-2)
f_gw = 2.75 * u.mHz
f_orb = f_gw / 2
strain = 1.8e-22
```

```
[ ]: chirp_mass = chirp_mass_from_chirp(fdot, f_orb)
print(chirp_mass.to(u.Msun))
```

11.018061108388425 solMass

```
[ ]: distance = distance_from_strain_chirp_mass(strain, chirp_mass, f_gw)
      print(distance.to(u.kpc))
```

894.466025780429 kpc

Based on the chirp and the observed frequency, the chirp mass suggests that the source is a binary black hole!

2 pts off for incorrect chirp mass

2 pts off for incorrect distance

2 pts off for incorrect interpretation